THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A quadrilateral is cyclic when its opposite angles are: 1

(a) complementary **(b)** supplementary **(c)** equal **(d)** both 90° only

2. A chord is at distance 20 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 30 **(b)** 32 **(c)** 29 **(d)** 31

3. The radius of a circle is 26 cm and a chord is 20 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 24 **(b)** 25 **(c)** 26 **(d)** 23

4. The chord AB subtends 136° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 58° **(b)** 88° **(c)** 78° **(d)** 68°

5. The chord AB subtends 144° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 128° **(b)** 108° **(c)** 118° **(d)** 98°

6. In a cyclic quadrilateral $ABCD$, $\angle A = 97^\circ$ and the side BC is produced to E. Then $\angle DCE =$ 1

(a) 87° **(b)** 117° **(c)** 97° **(d)** 107°

7. In triangle OAB, $OA = OB$ and $\angle AOB = 140^\circ$ (O is the centre). Then $\angle OAB =$ 1

(a) 20° **(b)** 40° **(c)** 30° **(d)** 10°

8. Angles in the same segment of a circle are: 1

(a) supplementary **(b)** complementary

(c) equal **(d)** double of each other

9. A chord is at distance 10 cm from the centre of a circle of radius 26 cm. Its length (in cm) is: 1

(a) 47 **(b)** 49 **(c)** 48 **(d)** 50

10. The radius of a circle is 29 cm and a chord is 42 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 20 **(b)** 21 **(c)** 22 **(d)** 19

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. $ABCD$ is a cyclic quadrilateral and BC is produced to E. If $\angle A = 91^\circ$, find $\angle DCE$. 2

12. In a circle of radius 29 cm, a chord is 21 cm from the centre. Find its length. 2

13. A chord AB of a circle with centre O subtends an angle of 130° at O. Find the angle it subtends at a point C on the minor arc. 2

14. $ABCD$ is a cyclic quadrilateral and BC is produced to E. If $\angle A = 109^\circ$, find $\angle DCE$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In triangle OAB with O the centre of a circle, $\angle AOB = 150^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. 3

16. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 12)^\circ$ and $\angle C = (3x - 7)^\circ$. Find x and $\angle A$. 3

17. A chord of length 48 cm is 7 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. 3

OR

In triangle OAB with O the centre of a circle, $\angle AOB = 50^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. 3

18. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 8)^\circ$ and $\angle C = (3x - 3)^\circ$. Find x and $\angle A$. 3

OR

A chord of length 14 cm is 24 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two parallel chords of lengths 12 cm and 16 cm lie on the same side of the centre of a circle of radius 10 cm. Find the distance between them. 5

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 97^\circ$, $\angle B = 64^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. 5

20. Two parallel chords of lengths 6 cm and 8 cm lie on the same side of the centre of a circle of radius 5 cm. Find the distance between them. 5

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 112^\circ$, $\angle B = 76^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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